

Iris Flower Classification

Data set used: Iris.csv

This project uses the Iris dataset (Iris.csv) and focuses on building classification models using Scikit-learn. The goal is to classify iris flowers into three species: Setosa, Versicolor, and Virginica based on sepal and petal measurements.

Multiple machine learning algorithms are applied and evaluated to understand their performance on a standard classification problem. The algorithms used include Support Vector Machine, Logistic Regression, Naive Bayes, Decision Tree, and Random Forest.

The primary Scikit-learn modules used in this project are:

- **sklearn.svm:** For Support Vector Machine (SVC)
- **sklearn.linear_model:** For Logistic Regression
- **sklearn.naive_bayes:** For Naive Bayes classifier
- **sklearn.tree:** For Decision Tree classifier
- **sklearn.ensemble:** For Random Forest classifier
- **sklearn.model_selection:** For train-test splitting
- **sklearn.metrics:** For accuracy evaluation

Project Objectives:

- To classify iris flower species using machine learning algorithms.
- To implement multiple Scikit-learn classification models.
- To compare performance of different algorithms.
- To understand supervised classification using real-world datasets.

Machine Learning Algorithms:

1. Support Vector Machine (SVM)

Definition:

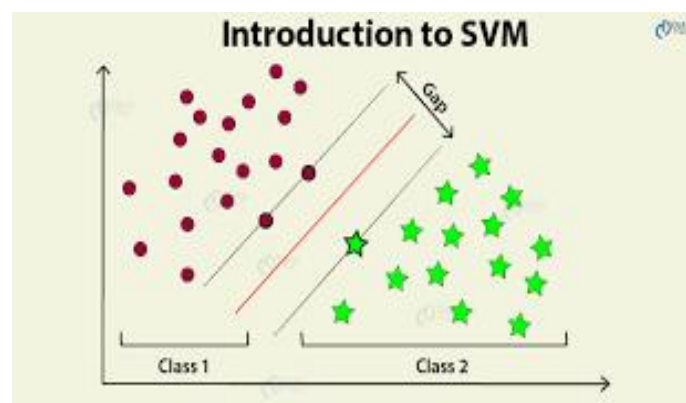
Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification. It finds the optimal boundary (hyperplane) that separates different classes with the maximum margin.

Use

- Used for classification problems with clear margins
- Works well in high-dimensional data
- Effective when classes are separable

Key Point

- Uses distance → requires feature scaling



2. Logistic Regression

Definition

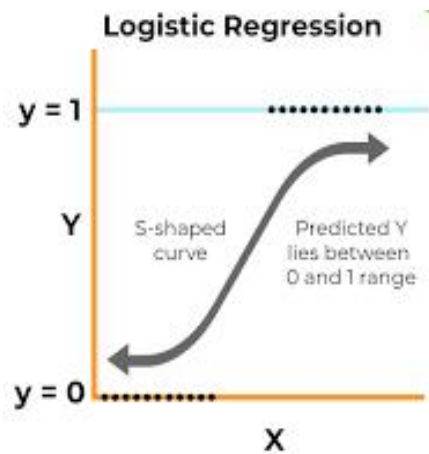
Logistic Regression is a classification algorithm that predicts the probability of a data point belonging to a particular class using a logistic (sigmoid) function.

Use

- Binary and multi-class classification
- Simple and fast model
- Works well with linearly separable data

Key Point

- Outputs probability, then converts to class .



3.Naive Bayes

Definition

Naive Bayes is a probabilistic classifier based on Bayes' Theorem, assuming that all features are independent of each other.

Use

- Text classification (spam detection, sentiment analysis)
- Works well with large datasets
- Fast and efficient

Key Point

- Assumes feature independence (not always realistic)

4.Decision Tree

Definition

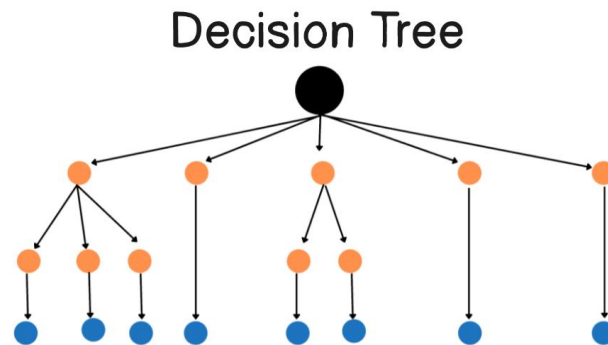
Decision Tree is a supervised learning algorithm that splits data into branches based on feature values, forming a tree-like structure for decision making.

Use

- Easy to understand and interpret
- Handles both classification and regression
- Works well with non-linear data

Key Point

- Can overfit easily



5. Random Forest

Definition

Random Forest is an ensemble learning method that builds multiple decision trees and combines their outputs to improve accuracy and reduce overfitting.

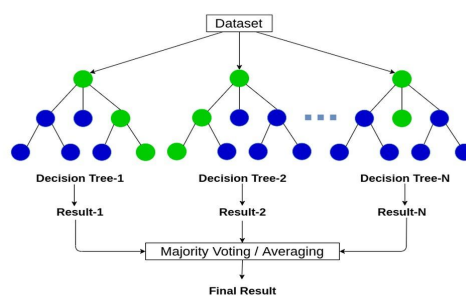
Use

- High accuracy models
- Handles large datasets
- Reduces overfitting compared to decision trees
-

Key Point

- Uses multiple trees (bagging technique)

Random Forest



Conclusion

In this project, multiple machine learning algorithms such as SVM, Logistic Regression, Naive Bayes, Decision Tree, and Random Forest were applied to classify iris flowers into three species. Each model showed different levels of performance based on how it learned patterns from sepal and petal measurements. Overall, the models were able to classify the species effectively, demonstrating that machine learning algorithms can accurately solve real-world classification problems.

What You Have learned

I learned how to apply different machine learning algorithms using Scikit-learn for classification tasks. I understood how features like sepal and petal measurements are used to predict flower species. I also learned how to split data into training and testing sets and evaluate models using accuracy score. Additionally, I gained knowledge about how different algorithms behave and how their performance can vary on the same dataset.